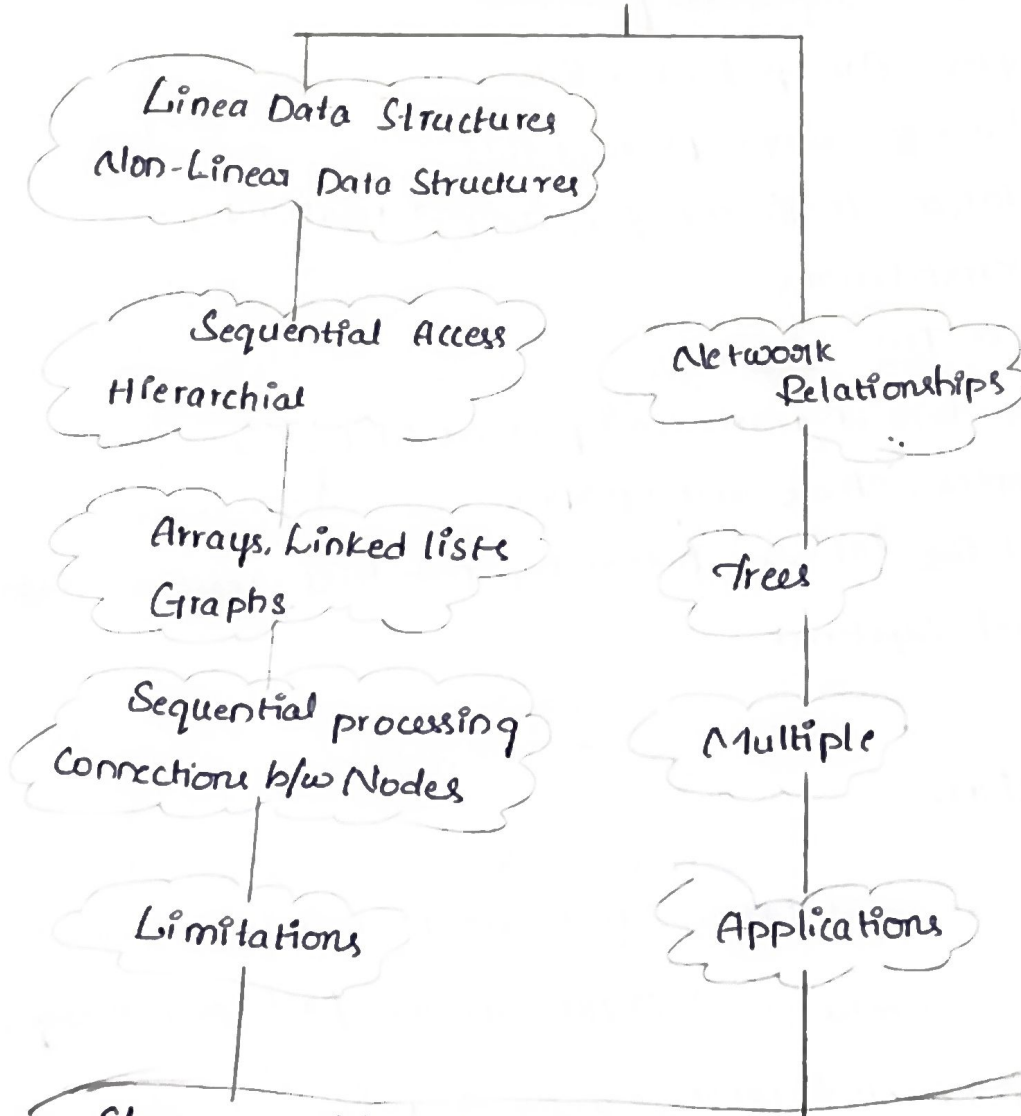


Assessment-1

E-Commerce Platform - Data Structure.

E-Commerce Platform.



Slow searching in large data product categories Difficult relationships Recommendation System product Search. Navigation.

Analysis:-

- Linear Data Structures

- Store data Sequentially
- Examples: Arrays, Linked lists
- Suitable for simple product lists.
- Limitation: Inefficient for complex relationships and recommendations

- Non-linear Data Structures

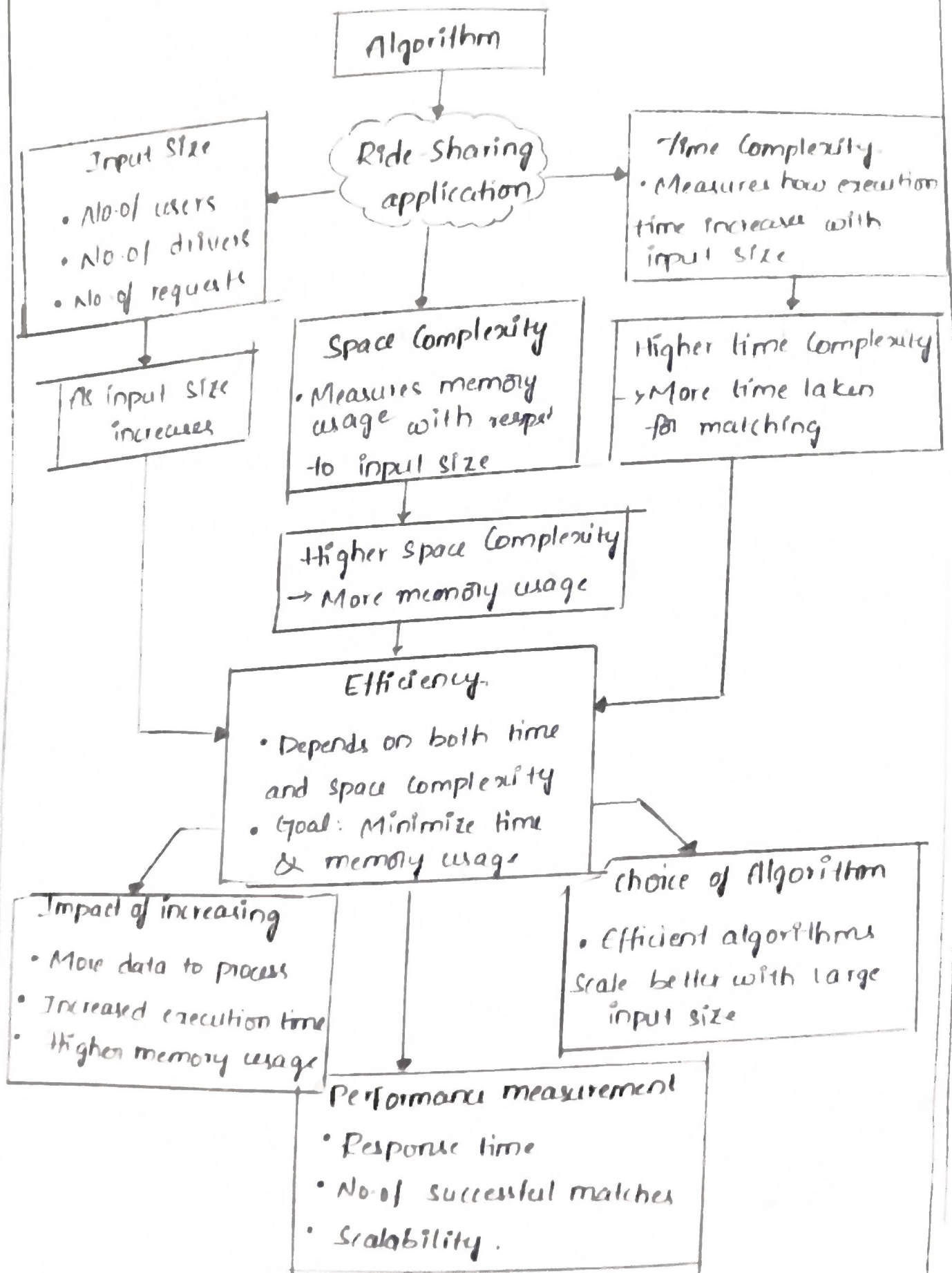
- Store data hierarchically or as Networks
- Examples: Trees and Graphs
- Used for category hierarchy, product recommendations and Search Systems.

Justification:-

Non linear datastructures are more efficient because they support hierarchical organization, fast searching, and interconnected relationships required in modern e-commerce platforms

- Search Efficiency: If an ecommerce platform with 10^7 products used a linear limited list for searching, finding product could require up to 10^7 operations. A balanced non-linear tree reduces this maximum boundary to $\log_2(10^7) = 24$ operations

Ride-sharing Application - Algorithm Efficiency



• Relational Density: Recommendation system requires calculating highly interconnected network clusters. A linear structure trying to map these relationships would require an adjacency matrix of size $n \times n$. (leading to catastrophic spatial wastage ($O(n^2)$ space complexity) due to data sparsity. Graph structures store only active relationships optimizing both space and relational traversal speed

Therefore, while linear structures remain ideal for simple, predictable pipelines - the core revenue-driving engine - Search and personalization depend entirely on the mathematical efficiency of non-linear structures